Introduction to Cryptography

Lecture 39: Message Authentication Techniques

Dr. Siddhant Thapliyal

Department of Computer Science and Engineering
Graphic Era (Deemed to be University), Dehradun

Learning Outcomes

- Explain message authentication goals.
- Distinguish integrity, origin authentication and confidentiality.
- Compare MACs, hashes and digital signatures.
- Describe replay protection and authenticated encryption.

What Is Message Authentication?

- Check that a received message came from an expected source.
- Detect unauthorized changes to message content.
- Often also require freshness to prevent replay.
- Authentication does not automatically hide the message.

Threats to Messages

- Masquerade: attacker impersonates a sender.
- Content modification: data altered in transit.
- Sequence modification: messages inserted or reordered.
- Replay: valid old messages resent maliciously.

Three Main Mechanisms

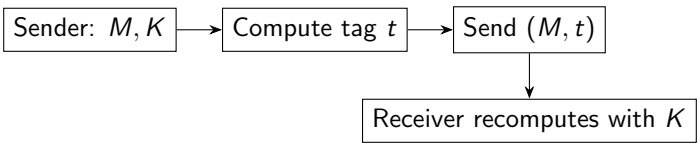
- Message Authentication Code (MAC).
- Cryptographic hash combined with a suitable authentication mechanism.
- Digital signature using a private/public key pair.

Hash vs. MAC

Hash	MAC
No secret key required	Shared secret key required
Detects accidental changes when trusted reference exists	Authenticates message under shared key
Publicly computable	Not computable without key

MAC Workflow

$$t = \mathrm{MAC}_K(M)$$



HMAC

HMAC combines a cryptographic hash with a secret key.

$$\mathrm{HMAC}_K(M) = H((K \oplus opad) \,\|\, H((K \oplus ipad) \,\|\, M))$$

ipad and *opad* are fixed pads; $\|$ denotes concatenation.

Digital Signatures

$$\sigma = \mathsf{Sign}_{SK}(M)$$

$$\mathsf{Verify}_{PK}(M, \sigma) \in \{\mathsf{accept}, \mathsf{reject}\}$$

Signatures enable public verification, unlike symmetric MACs.

MAC vs. Signature

MAC	Digital signature
Shared secret key	Private signing/public verification keys
Fast, symmetric	Asymmetric operation
Either key holder could create tag	Verifier need not know signing key

Replay Protection

- Attach sequence numbers, timestamps, or unique nonces.
- Bind freshness fields to the authenticated data.
- Receiver checks freshness and rejects repeated values.
- A valid MAC alone cannot prevent replay of a previously valid message.

Authenticated Encryption

- AEAD provides confidentiality and integrity/authentication together.
- Examples include AES-GCM and ChaCha20-Poly1305.
- Nonces must follow each algorithm extquotesingle s usage requirements.
- Authenticate protocol metadata as associated data where appropriate.

Illustrative Python HMAC Demo

Python 3

```
import hmac, hashlib
key = b"shared-secret"
msg = b"Transfer 100"
tag = hmac.new(key, msg, hashlib.sha256).digest()
print(hmac.compare_digest(tag,
 hmac.new(key, msg, hashlib.sha256).digest()))
```

Class Exercise

1. Why is an ordinary hash insufficient to authenticate a sender?
2. How does a nonce prevent replay?
3. Can a MAC provide public verifiability?
4. Which technique provides both secrecy and authenticity?

Summary

Authentication $\neq$ Confidentiality

Use MACs, signatures, freshness controls, and AEAD as appropriate.

Your Feedback Matters!

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